

SAGE: Slot-Aware Grounded Evidence Acquisition for LitTraceQA

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Abstract

Literature-grounded question answering requires a system to retrieve the right papers, locate answer-bearing content in heterogeneous PDFs, and return an answer whose evidence locators can be restored from the retrieved documents. We describe SAGE, an evidence-first system for LitTraceQA that separates these dependencies. A query-only evidence plan converts a question and its public answer contract into typed evidence obligations. The system retrieves candidate papers from official metadata, acquires and parses their PDFs into provenance-preserving text and object records, reranks record-aware text and image projections, and selects a bounded evidence package with explicit slots, modality, and cross-paper coverage priorities. A multimodal generator then answers with the fixed package, while deterministic normalization permits only candidate paper IDs and locators recoverable from selected parser records. Overall, SAGE adopts an evidence-first design that separates paper retrieval, fine-grained evidence acquisition, set-level selection, and grounded answer generation while preserving provenance throughout the pipeline. On the official evaluation, the system performs strongly in paper retrieval and multiple-choice answering, whereas fine-grained evidence grounding and structured table prediction remain more challenging.

1 Introduction

Scientific literature grows faster than researchers can manually locate, read, and reconcile relevant papers. Retrieval-augmented language models condition generation on external evidence: REALM incorporates retrieval into language-model pretraining (Guu et al., 2020), while RAG formalizes retrieval-conditioned generation for knowledge-intensive tasks (Lewis et al., 2020). LitTraceQA,

however, requires more than retrieving a topical passage. Questions may ask for a table value, a figure-supported comparison, an equation, or a citation record spread across several scientific papers. This creates a heterogeneous, cross-modal evidence retrieval problem.

Long-document QA benchmarks such as QASPER show that research questions are often grounded in lengthy documents rather than self-contained passages (Dasigi et al., 2021). In this setting, a retrieved item can be topically relevant yet fail to supply the requested value or condition. Conversely, individually useful items may not collectively cover a multi-paper question. Citation-aware generation work similarly shows that answer quality depends on whether cited evidence actually supports generated claims (Gao et al., 2023b). We therefore treat LitTraceQA as an evidence-acquisition problem rather than passage retrieval followed by generation. A successful system must not only identify useful evidence, but also assemble complementary evidence across sources and preserve a traceable path from retrieved content to the final output.

SAGE is an end-to-end system built around this principle. It does not propose a new base retriever or a new multimodal model. Instead, it makes the interfaces between paper retrieval, PDF parsing, evidence selection, answer generation, and provenance restoration explicit. Inspired by task decomposition and plan-first prompting (Khot et al., 2023; Wang et al., 2023), a query-only plan is frozen before paper retrieval; typed records retain the paper, page, source type, and object locator needed for the final task format; and selection is separated from item-level reranking so that the context package can cover several obligations under a bounded budget.

Our system contribution is threefold:

- **Query-only evidence routing.** We turn

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a question and its public answer contract into explicit evidence slots that guide task-family routing, reranking, and package selection without predicting an answer or reading candidate-paper content.

- **Provenance-preserving multimodal evidence packages.** We represent parsed PDFs as typed records and package objects with their local explanatory context, retaining a reversible path from a selected item to an admissible evidence locator.
- **Evidence-package grounded answering.** We distinguish item-level utility from set-level coverage, select a bounded package through heterogeneous routes, and project generated output onto the candidate papers, selected evidence, and public answer schema.

The remainder of this paper describes the task, SAGE methodology, experimental results, and limitations. Code and reproducibility materials are available at <https://github.com/constantine617/DKE-SAGE>.

2 Related Work

Retrieval-augmented generation couples external evidence with answer generation (Lewis et al., 2020); Fusion-in-Decoder extends this idea by aggregating multiple retrieved passages (Izacard and Grave, 2021). SAGE differs in its system boundary: it must retrieve canonical papers, preserve submission-compatible evidence locators, and handle heterogeneous PDF objects rather than treating all evidence as interchangeable passages.

Scientific-document QA requires document-level reasoning and structural evidence. QASPER provides long research papers as the source of answerable questions (Dasigi et al., 2021), while LitTraceQA requires paper identifiers, locators, and structured answers in addition to answer content (Liu et al., 2026). Our use of Docling follows the need for structured PDF transcription (Auer et al., 2024), but the contribution is not a new parser; it is the provenance-preserving interface between parsed records and evidence-grounded output.

Finally, the system combines heterogeneous routes through reciprocal-rank fusion (Cormack et al., 2009) and optionally derives multi-paper lexical routes using HyDE (Gao et al., 2023a). These mechanisms expose complementary candidate views at the package-selection boundary. They

are used here as system components, not presented as new retrieval algorithms.

Planning methods such as ReAct (Yao et al., 2023), least-to-most prompting (Zhou et al., 2023), and plan-and-solve prompting (Wang et al., 2023) decompose a task before or during reasoning. SAGE adopts decomposition only to create an auditable retrieval interface: its plan is schema-validated, frozen before candidate-paper content is read, and cannot contain answer values or locators.

3 Task and System Boundary

LitTraceQA is one of the two GroundLM 2026 shared tasks, together with GoldenViewVQA (Wang et al., 2026a), which evaluates view-level visual evidence identification. This work participates in LitTraceQA (Liu et al., 2026), a benchmark for multi-stage grounding and verification over scientific literature. The overall shared-task setup and evaluation are summarized by Wang et al. (2026b).

Given a question q , its public answer contract τ , and an official metadata collection $\mathcal{D}$, the system must return a prediction

$$y = (P, E, A),$$

where $P \subseteq \mathcal{D}$ is a set of canonical paper identifiers, E is a set of evidence locators, and A is an answer conforming to τ . The public contract supplies the requested answer structure, including multiple-choice options or a table schema when applicable.

More concretely, the public input contains the question, the answer-format contract, and the official metadata records used for candidate-paper retrieval. The system then obtains the corresponding source PDFs and must emit three benchmark-facing objects: canonical paper IDs, evidence entries with a permitted source type and locator, and an answer conforming to the requested multiple-choice or table schema.

The task naturally has three dependent stages:

$$p(P \mid q, \mathcal{D}), p(E \mid q, P, \tau), p(A \mid q, E, \tau).$$

This factorization is not a claim that the implementation trains three probabilistic models. It exposes a failure dependency: if a required paper is absent from the candidate set, its evidence is unavailable downstream; if an answer-bearing object is not parsed or selected, generation cannot ground the requested value. LitTraceQA requires canonical paper IDs, coarse evidence locators, and

a multiple-choice or schema-constrained table answer (Liu et al., 2026). Multi-paper questions additionally require evidence from complementary sources.

4 Methodology

4.1 System Overview

Figure 1 summarizes the end-to-end execution path of SAGE. Given a question q and its public answer contract τ , the system first constructs a query-only evidence plan that specifies the information to be retrieved and the routing signals used by downstream components. The plan is frozen before candidate-paper content is inspected. Candidate papers are then retrieved from the official metadata collection, and their PDFs are parsed into typed records that preserve paper, page, source-type, and object-level provenance.

Text records are reorganized into retrieval units in relatively uniform chunks, while other objects preserve their records as units. Then these units exposed through query-conditioned, slot-conditioned, source-type, object, and paper-local routes. A multimodal reranker estimates the utility of individual evidence units. The resulting rankings are fused before a coverage-aware selector constructs a bounded evidence package that accounts for plan-derived requirements, requested modalities, and cross-paper coverage. Once the package is selected, the evidence boundary is fixed: the generator answers only from the selected evidence, and deterministic output projection maps its structured response to admissible paper identifiers, evidence locators, and the required answer schema.

Two signals persist across these stages. The evidence plan acts as a forward control interface for retrieval and selection, while provenance established during PDF parsing is preserved through final output projection.

4.2 Query-Only Evidence Planning and Routing

Rather than treating q as a single retrieval string, SAGE first constructs a frozen evidence plan

$$\mathcal{S}(q, \tau) = \{s_1, \dots, s_m\}, s_j = (r_j, o_j, T_j, N_j, C_j),$$

where r_j denotes the evidence role, o_j the required operation, T_j the preferred source types, N_j the named entities, and C_j measurable conditions

such as datasets, metrics, settings, or configurations.

The planner receives only q and τ . In addition to these slot-level requirements, it predicts query-level routing attributes, including expected paper scope and primary evidence type. Paper scope determines the corresponding candidate-paper and evidence budgets, while evidence-type predictions provide weak preferences to matching source-type and object routes. The planner is explicitly prevented from supplying answer values, paper identities, pages, or evidence locators.

The plan is schema-validated and frozen before candidate-paper content is read. It is subsequently reused as a control interface for candidate retrieval, slot-conditioned evidence routing, and coverage-aware selection. It therefore specifies what evidence should be sought rather than attempting to answer the question itself.

4.3 Candidate paper Reachability

Candidate retrieval constructs a bounded paper set

$$C_P(q) \subseteq \mathcal{D}.$$

If a required paper p^* is not in $C_P(q)$, evidence unique to that paper cannot be recovered by later stages:

$$p^* \notin C_P(q) \Rightarrow \mathcal{E}(p^*) \cap \mathcal{E}(C_P(q)) = \emptyset.$$

The implementation uses hybrid-alias lexical retrieval over normalized official metadata, with title mentions, aliases, acronyms, venue, and year signals. BM25 provides the lexical retrieval primitive (Robertson and Zaragoza, 2009). For questions routed as multi-paper, query decomposition contributes bounded subquery views before the candidate list is merged. Each candidate PDF is then availability-audited before parsing, so a missing source is not silently conflated with an extraction failure. The evidence plan does not assign a paper to a slot at this stage; paper-specific slot routes are created only when a candidate has a unique, title-supported match. This separation prevents an early plan from becoming an unsupported paper prediction.

4.4 Provenance-Preserving Evidence Representation

For every reachable paper, the system downloads the PDF and converts it into canonical records

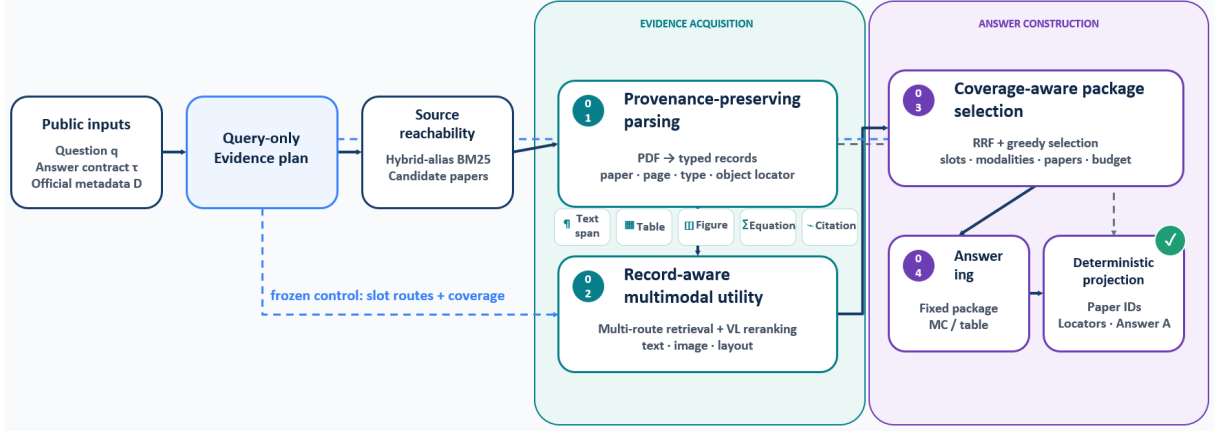


Figure 1: SAGE system architecture. Solid arrows show the executed pipeline; blue dashed arrows reuse the query-only plan for routing and coverage, while gray dashed arrows preserve provenance through output projection.

rather than flat chunks. A record is represented as

$$e_i = (p_i, l_i, t_i, x_i, m_i, v_i),$$

where p_i is the paper ID, l_i is the page/object locator, t_i is the official source type, x_i is text, m_i is structural metadata, and v_i is an optional visual crop. The source types are text_span, table, figure, equation_algorithm, and citation_context.

Docling with OCR is the production transcription backend (Auer et al., 2024). It supplies document structure, while the standardizer retains source types and locators before any answer is generated. Object records are packaged with captions and bounded neighboring text; table, figure, and equation/algorithm crops supplement their symbolic representation. This design preserves the granularity required by the benchmark: a table is not reduced to unrelated prose, and a final locator is restored from a parser record rather than generated freely.

4.5 Record-Aware Multimodal Reranking

Canonical records are reorganized into retrieval units before fine-grained ranking. Text records may be packed into bounded local units, while tables, figures, equations/algorithms, and citation records remain atomic objects accompanied by their local explanatory context. Each unit retains the provenance of its parent record.

SAGE constructs multiple candidate rankings from the original query, plan-derived slots, source-type and object cues, and paper-local context. BM25 is used to retrieve and rank units within these

lexical routes, and the union of their candidates is then passed to the Qwen multimodal reranker for fine-grained scoring. For visually grounded units, the reranker considers both textual and visual representations, while lightweight structural signals provide additional preference for explicitly requested source types or object labels.

These rankings estimate the usefulness of individual evidence units for the question and its plan-derived requirements. Reranking remains an item-level operation: a highly ranked unit may satisfy one requirement without providing the complete evidence set required by the question. This motivates the separate set-level selection stage that follows.

4.6 Coverage-Aware Evidence Package Construction

The selector receives a pool $\mathcal{C}_E$ of record-aware packages. It does not optimize an unobserved probability of sufficiency. Instead, the implementation executes a deterministic, budgeted selection procedure:

$$E^* = \text{GreedySelect}(\mathcal{C}_E; \mathcal{R}_{\text{slot}}, \mathcal{M}, \mathcal{P}, B),$$

where $\mathcal{R}_{\text{slot}}$ is the set of slot routes, $\mathcal{M}$ denotes requested source-type coverage, $\mathcal{P}$ denotes paper-diversity requirements for multi-paper questions, and B is the package and context budget. The formulation mirrors the actual priority schedule rather than introducing unimplemented learned weights.

The selector builds heterogeneous rankings from the original query, source-type and object routes, plan-derived slot routes, optional

multi-paper HyDE claims (Gao et al., 2023a), paper-conditioned claims, paper-local lexical routes, Qwen text/image ranks, and layout ranks. Reciprocal-rank fusion combines these lists without requiring their scores to be calibrated (Cormack et al., 2009). It first attempts to cover plan routes, explicit objects, requested modalities, and distinct papers; it then adds page-diverse high-utility packages until the package or character budget is reached. Adaptive stopping is allowed only after the minimum package budget and requested coverage are satisfied. The resulting selector is a transparent rule-based coverage procedure, not a generic submodular optimizer.

The distinction is central: reranking estimates the utility of individual evidence units, whereas selection constructs a bounded evidence package. A package remains context-atomic: if it contains an object and its explanatory records, it is included whole or omitted, rather than truncated into an ungrounded fragment.

4.7 Grounded Answer Generation

Once E^* has been selected, the evidence boundary is fixed. The multimodal generator receives the question, the public answer contract, and only the selected evidence package. Depending on the contract, it produces either a multiple-choice prediction or a schema-constrained table. Selected tables, figures, and equations/algorithms may be supplied through both their symbolic records and visual crops.

Generation cannot initiate another paper-retrieval or evidence-discovery step. This keeps answer generation downstream of evidence acquisition and prevents newly generated content from being treated as retrieved evidence.

4.8 Provenance-Constrained Output Projection

The generator produces an internal structured response, but benchmark-facing paper IDs and evidence locators are restored only from retrieved provenance. Let $\mathcal{L}(E^*)$ denote the set of admissible locators recoverable from the parser records represented in the selected evidence package. The final prediction $y = (P, E, A)$ must satisfy $P \subseteq C_P$, $E \subseteq \mathcal{L}(E^*)$, $A \in \mathcal{A}_\tau$.

The projection stage canonicalizes paper identifiers, resolves evidence references to selected parser records, and normalizes multiple-choice or table outputs against the public contract. It cannot

Component	Configuration
Evidence planner	DeepSeek-V4-Flash; temperature 0
Answer generator	Qwen3-VL-32B-Instruct; temperature 0
Multimodal reranker	Qwen3-VL-Reranker-8B
Candidate-paper retrieval	Hybrid-alias BM25; top-5 single / top-12 multi
Page routing	5 pages/candidate single / 3 multi
PDF transcription	Docling + OCR with PyMuPDF provenance; 180 DPI crops
Evidence selection budget	12 units single / 36 multi
Route fusion	Reciprocal-rank fusion, $k = 60$
Retrieval subquery budget	top-4 per subquery
Auxiliary route generation	DeepSeek-V4-Flash; up to 12 papers

Table 1: Configuration of the reported SAGE system. Single- and multi-paper budgets are determined by the query-visible routing signal produced before evidence retrieval.

retrieve additional sources or introduce new evidence after generation. This guarantees structural and provenance validity, but not semantic correctness.

5 Experiments and Result Analysis

The official LitTraceQA submission was made under team DKE; SAGE denotes the submitted system described in this paper.

5.1 Experimental Setup

Table 1 summarizes the frozen configuration associated with the reported SAGE run. We perform no task-specific training or fine-tuning; all neural components are used as pretrained inference-time models. The evidence planner and auxiliary route generation use DeepSeek-V4-Flash, while answer generation and multimodal reranking use separate Qwen models.

Candidate-paper and page budgets bound the evidence space before fine-grained reranking. The evidence selection budget bounds the number of packages admitted to the final context. Rankings produced by the heterogeneous evidence routes described in Section 4 are fused using reciprocal-rank fusion (RRF) with $k = 60$.

For development-side analysis, we additionally evaluate SAGE on the released 55-query validation set using the official evaluator and validation gold. To match the hidden-test input format, multiple-choice option lists are restored for the 41 validation queries evaluated as multiple choice; gold answers and gold evidence are not provided to the

Metric	Clean baseline	SAGE
Paper F1	.7199	.9915
Evidence F1	.3179	.7643
MC accuracy	.7200	1.0000
Table Row F1	.4469	.5927
Table Cell Macro	.1349	.3413
Table Cell Micro	.1379	.3793
Composite	.5516	.8002

Table 2: Official hidden-test results.

Metric	Validation
Paper Precision / Recall / F1	1.0000 / 1.0000 / 1.0000
Evidence Precision / Recall / F1	.5874 / .5525 / .5612
MC accuracy	.6585 (27/41)
Freeform EM	.6923 (18/26)
Table Row F1	.5247
Table Cell Macro / Micro	.4520 / .3333

Table 3: Official-evaluator performance of SAGE on the 55-query validation set. Multiple-choice and freeform scores are reported over the instances evaluated for the corresponding answer type.

model. Validation results are used only for diagnostic analysis and are not used to construct the reported hidden-test submission.

5.2 Main Results

Table 2 compares SAGE with the clean baseline on the official hidden test, where the clean baseline consists of paper retrieval without an evidence plan and multi-route, record-reranking, and generation. In general, the architecture of the clean baseline is close to the Naive RAG(Lewis et al., 2020). All reported values are returned by the organizer’s evaluator.

SAGE improves over the coverage-first baseline on every reported task component. Paper F1 increases from 0.7199 to 0.9915 and Evidence F1 from 0.3179 to 0.7643, while SAGE achieves perfect multiple-choice accuracy. Structured table prediction also improves, although its cell-level scores remain substantially below those of paper retrieval and multiple-choice performance. The comparison is between two complete system configurations and should therefore not be interpreted as a controlled component ablation.

We also run the SAGE pipeline on the released 55-query validation set. Table 3 reports the corresponding official-evaluator results.

The validation results suggest that the main difficulty lies after source identification rather than in paper retrieval itself. SAGE retrieves all required papers on the 55-query split, yet only about half of the gold evidence is recovered in the final pre-

diction. A similar pattern appears in structured answering: table row structure is recovered more reliably than exact cell values, suggesting that identifying the relevant comparison structure is easier than extracting and composing all required values. Multiple-choice and freeform performance are moderate on this split, showing that successful source retrieval therefore does not guarantee complete evidence attribution or correct answer generation.

Comparison with the 71-query hidden test shows a stable strength at the paper-retrieval stage but substantially larger variation downstream. Paper retrieval is nearly saturated on both splits (1.0000 F1 on validation and 0.9915 on the hidden test), whereas evidence and answer metrics differ more noticeably: Evidence F1 increases from 0.5612 to 0.7643 and multiple-choice accuracy from 0.6585 to 1.0000 on the hidden test. Table performance is more mixed, with a higher Row F1 and Cell Micro score on the hidden test but a lower Cell Macro score. Because the two splits differ in question and answer-type composition, these differences should not be interpreted as generalization gains or degradation. Instead, the consistent pattern across both evaluations is that candidate-paper retrieval is comparatively robust, while fine-grained evidence grounding and exact structured answer composition remain more sensitive to the individual questions.

5.3 Failure Analysis

The archived evaluation evidence exposes three identifiable failure modes in the final system.

Residual source-reachability failure. Paper recall reaches 0.9894 rather than 1.0. The archived evaluation analysis corresponds to one required paper being missed for one question. Although rare, this error remains an absolute upstream failure: evidence unique to an unretrieved paper is inaccessible to all downstream stages.

Answer correctness without correct evidence grounding. Two multiple-choice questions obtain the correct final answer while their submitted evidence sets contain no evidence true positives under the examined locator formulations. These cases demonstrate that answer accuracy and evidence accuracy are not equivalent. Correct answer generation can coexist with an incorrect or incomplete evidence submission, motivating separate evaluation of answer prediction and evidence grounding.

Structured table composition. Table prediction remains the weakest answer component. Among 87 evaluated non-key gold cells, 54 are incorrect, and fourteen examined candidate row keys are entirely non-matching. At the same time, the scored submissions satisfy the official output validator. The observed errors therefore cannot be explained by schema invalidity alone; they are concentrated in exact row-key and cell-value composition. This is consistent with the difference between the Row F1 of 0.5927 and the lower cell-level macro and micro scores of 0.3413 and 0.3793.

Overall, the available evidence places most remaining error after source identification. SAGE can usually reach the relevant papers and produce a valid submission, but reliable fine-grained evidence attribution and exact structured value composition remain the principal unresolved challenges.

6 Limitations

The system cannot recover evidence from a required paper that is absent from the metadata candidate set, unavailable as a PDF, or incorrectly parsed. Its deterministic projection prevents invented locators, but does not verify that the generated answer is semantically entailed by the selected evidence. The evidence plan and multimodal reranker are also external-model calls, so their behavior can change when a provider changes a deployed model. Finally, the package budget necessarily trades completeness against prompt capacity: a multi-paper question may require more mutually complementary evidence than a bounded context can contain. These are system limitations rather than claims resolved by provenance normalization.

7 Ethical Considerations

SAGE processes only organizer-released benchmark inputs, public paper metadata, and source PDFs obtained from permitted public routes. It does not collect private user data or use hidden benchmark labels during inference. Nevertheless, retrieval, PDF parsing, and remote language-model outputs may be incomplete or incorrect. Provenance-constrained projection prevents unsupported paper IDs and locators from entering the final submission, but it does not guarantee semantic correctness. The system should therefore support, rather than replace, human verification when its outputs are used for scientific interpretation. Source PDFs remain subject to their original licenses and

access conditions; the system records locators and derived predictions rather than redistributing source documents.

8 Conclusion

SAGE organizes LitTraceQA around explicit dependencies that are frequently conflated in standard retrieval pipelines. Candidate retrieval determines whether a source is reachable; slot-conditioned reranking estimates the utility of individual evidence packages; coverage-aware selection determines whether a bounded set can meet the question’s requirements; and output projection enforces a reversible provenance and schema boundary. The system deliberately does not equate any one of these properties with semantic correctness. This separation provides a concrete basis for subsequent evaluation of where LitTraceQA succeeds or fails, especially when evidence is heterogeneous or distributed across papers.

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A External Resources and Generated Artifacts

This appendix summarizes the external resources, pretrained models, software dependencies, and generated artifacts used by SAGE. The reported system uses pretrained components only and performs no task-specific training or fine-tuning.

A.1 Benchmark Data and Source Documents

For the reported hidden-test submission, the system uses only the organizer-released LitTraceQA test inputs, the official paper metadata, the public answer-format contract, and the official submission validator. Candidate source documents are obtained from the document URLs associated with the official metadata and constitute the evidence corpus for inference. No external document collection, web-search corpus, manually curated paper list, or additional supervision is introduced.

The released validation split and its gold annotations are used separately for development-side evaluation and diagnostic analysis. Multiple-choice option text and table schemas are treated as part of the public answer contract rather than as answer supervision.

A.2 Pretrained Models

SAGE uses the following pretrained models through remote inference services:

- **DeepSeek-V4-Flash** for query-only evidence planning and auxiliary retrieval-route generation;
- **Qwen3-VL-Reranker-8B** for record-aware textual and visual evidence reranking; and

- **Qwen3-VL-32B-Instruct** for final evidence-conditioned answer generation.

All models are used without task-specific parameter updates. The reported configuration uses temperature 0 for the evidence planner and answer generator. OCR and document-layout components required by the PDF-processing pipeline may additionally load pretrained model weights during environment setup; these models are used only for document transcription and layout analysis.

A.3 Software and External Services

The reference implementation uses Python 3.12. Its principal software dependencies include Doctling 2.113.0 for structured PDF transcription, PyMuPDF 1.24.14 for page rendering and provenance-preserving crops, RapidOCR with ONNX Runtime for OCR execution, rank-bm25 0.2.2 for lexical retrieval, and jsonschema for structured-output validation.

Remote language-model and reranking calls are made through the SiliconFlow OpenAI-compatible API. Network access is also required to obtain source PDFs and, when necessary, pretrained dependencies for document processing. Because model inference is provided by external services, provider-side model or serving revisions may introduce nondeterminism even when temperature-zero decoding is used.

A.4 Generated Inference Artifacts

SAGE does not use synthetic training examples, synthetic answer labels, synthetic evidence labels, or a generated augmentation corpus. During inference, however, the system produces intermediate model-generated artifacts, including query-only evidence plans and optional auxiliary retrieval-route expansions. These artifacts are used only as routing and retrieval signals and are not reused as training data.

The final answer is also model-generated, but benchmark-facing paper IDs and evidence locators remain constrained by the retrieved candidate papers and provenance-preserving parser records. Generated planning or routing content therefore cannot bypass the provenance boundary established by the retrieval and parsing pipeline.

Figure 1 is a schematic illustration of the SAGE architecture constructed from the implemented system workflow. It does not depict a benchmark ex-

ample, source-paper content, or experimental measurements.

B Prompt Templates

The complete prompt templates used by SAGE, including evidence planning, auxiliary retrieval planning, multimodal reranking instructions, evidence-grounded answer generation, structured-output retries, and target JSON schemas, are released at <https://github.com/constantine617/DKE-SAGE>.

The repository also provides the corresponding configuration files and identifies conditional prompt branches separately from deterministic pipeline components.